

L Adaptive Control for Full-State Stabilization of an Inverted Pendulum on a Cart

1 Introduction

This document presents an L adaptive controller designed for robust full-state stabilization of a fourth-order inverted pendulum on a cart. The controller integrates a nominal LQR controller with an adaptive augmentation layer that compensates for modeling uncertainties and external disturbances. A state predictor, an adaptation law with a projection operator, and a low-pass filter work together to ensure both transient performance and long-term stability.

2 Controller Architecture

The overall L adaptive control scheme comprises the following components:

1. **Nominal LQR Controller:** Provides baseline stabilization by computing a gain K based on a state-space model. The LQR is tuned (via the Q and R matrices) to prioritize cart position stabilization.

2. **Adaptive Augmentation Layer:**

- **State Predictor:** Estimates the system state $\hat{x}$ using nominal dynamics augmented by the adaptive estimate.

$$\dot{\hat{x}} = A\hat{x} + B(u + \hat{\sigma})$$

- **Adaptation Law:** Updates the adaptive estimate $\hat{\sigma}$ based on the prediction error $x - \hat{x}$. The update law is given by

$$\dot{\hat{\sigma}} = \Gamma B^T P (x - \hat{x}),$$

where Γ is the adaptation gain and P is the solution of the Lyapunov equation

$$A_{cl}^T P + P A_{cl} = -I.$$

- **Projection Operator:** Limits the adaptive estimate to a prescribed bound $\sigma_{\max}$ to avoid windup, ensuring

$$\hat{\sigma} \in [-\sigma_{\max}, \sigma_{\max}].$$

3. **Low-Pass Filter:** Filters the adaptive estimate to smooth the control action. Its dynamics are

$$\dot{z}_f = -\frac{1}{\tau}z_f + \frac{1}{\tau}\hat{\sigma},$$

where τ is the filter time constant.

3 Overall Control Law and Closed-Loop Dynamics

The complete control input is formed by combining the nominal LQR control with the filtered adaptive augmentation:

$$u = -Kx - z_f.$$

The closed-loop system incorporates the following dynamics:

- a. **Plant Dynamics:**

$$\dot{x} = Ax + Bu + Ed_{\text{ext}}(t),$$

where E is the disturbance input matrix and $d_{\text{ext}}(t)$ represents external disturbances.

- b. **Predictor Dynamics:**

$$\dot{\hat{x}} = A\hat{x} + B(u + \hat{\sigma}).$$

- c. **Adaptation Law (with Projection):**

$$\dot{\hat{\sigma}} = \Gamma B^T P (x - \hat{x}),$$

modified via a projection operator to ensure $\hat{\sigma}$ remains within $[-\sigma_{\max}, \sigma_{\max}]$.

- d. **Low-Pass Filter Dynamics:**

$$\dot{z}_f = -\frac{1}{\tau}z_f + \frac{1}{\tau}\hat{\sigma}.$$

These dynamics are combined into an augmented state vector:

$$z = \begin{bmatrix} x \\ \hat{x} \\ \hat{\sigma} \\ z_f \end{bmatrix},$$

which is used to simulate the overall closed-loop performance.

4 Tuning Parameters and Their Effects

The L adaptive controller performance is influenced by several key parameters:

- **Adaptation Gain (Γ):** A higher Γ accelerates adaptation, enhancing disturbance rejection but may induce overshoot or instability if too high. A lower Γ results in a more conservative adaptation.
- **Low-Pass Filter Time Constant (τ):** A smaller τ yields faster filtering, reducing high-frequency noise, but might delay the adaptive response. A larger τ smooths the response at the expense of slower adaptation.
- **Projection Bound ($\sigma_{\max}$):** This parameter limits the magnitude of the adaptive estimate. Tight bounds improve robustness and prevent windup, but if set too low, they may restrict the controller's ability to compensate for large uncertainties.
- **Nominal LQR Weights (Q_{lqr} and R_{lqr}):** These determine the baseline performance of the LQR controller. Emphasizing cart position by increasing the weight on x_1 (cart position) ensures that the primary stabilization objective is met.

By adjusting these parameters, the controller can be tuned to balance responsiveness, robustness, and long-term stability in the presence of model uncertainties and external disturbances.

5 Simulation and Performance Evaluation

Simulation of the augmented system over an extended period verifies:

- **Transient Response:** The time responses of the cart position and pendulum angle indicate how quickly and accurately the system is stabilized.
- **Control Effort:** Monitoring the control input u ensures that it remains within acceptable limits.
- **Adaptive Dynamics:** The evolution of the adaptive estimate $\hat{\sigma}$ and the filtered term z_f demonstrate the controller's ability to adjust for disturbances and uncertainties.

The simulation results provide insight into the effectiveness of the L adaptive control scheme in achieving robust full-state stabilization.

6 Summary

The L adaptive controller presented here combines a nominal LQR controller with an adaptive augmentation layer to achieve robust stabilization of an inverted pendulum on a cart. The controller features:

- A state predictor that estimates the plant dynamics.
- An adaptation law with a projection operator to prevent windup.
- A low-pass filter to smooth the adaptive contribution.

Careful tuning of the adaptation gain, filter time constant, and projection bound is essential to balance aggressive disturbance rejection with long-term stability. Simulation results demonstrate that the integrated control strategy effectively stabilizes both the cart position and the pendulum angle in the presence of disturbances and uncertainties.